🍽️ DIGESTIVE SYSTEM — FULL MIND MAP

Biology Ch. 7.ii · Human Anatomy & Physiology · Hinglish · SSC / BPSC / BSSC / Railway

📄 Topics in this sub-chapter:

1. Digestive Process & Taste Sensation
2. Buccal Cavity, Saliva & Starch Digestion
3. Stomach — HCl & Gastric Digestion
4. Enzymes — Nature, Function & Examples
5. Liver — Functions, Bile & Fat Digestion
6. Lipids, Lipase & Digestive Enzymes
7. Small Intestine, Appendix & Structural Order
8. Protein Digestion — Pepsin & Trypsin
9. Cellulose, Diastase & Carbohydrate End Products
10. Enzyme / Hormone Match-the-Following
11. Glycogen Storage, Fasting & Gallbladder
12. Pancreas & Intestinal Hormones

1. Digestive Process & Taste Sensation

- Human body mein digestion mouth se start hota hai aur source ke mutabik digestive process ka major hissa / completion small intestine mein hota hai. Small intestine pancreatic enzymes aur liver bile ki help se food breakdown continue karti hai. *[Q1]*
- Source five basic tongue tastes ko Sweet, Bitter, Salty, Umami aur Sour batata hai. “Spicy” aur “Pungent” ko actual taste nahi maana gaya; spicy sensation capsaicin se pain/heat signal ke roop mein explain ki gayi hai. *[Q2]*
- Source ke mutabik tongue ke sab parts tastes sense kar sakte hain, lekin posterior/back part bitterness ke liye especially sensitive hai. *[Q3]*

2. Buccal Cavity, Saliva & Starch Digestion

- Buccal cavity mein carbohydrate digestion start hota hai. Salivary amylase starch ko maltose/dextrin mein todna shuru karta hai. *[Q4]*
- Saliva dietary starch digestion mein help karti hai; source saliva ko about 99.5% water aur remaining fraction mein electrolytes, mucus, glycoproteins, enzymes (amylase) aur antibacterial compounds batata hai. *[Q5, Q6]*
- Amylase starch hydrolysis se glucose formation se associated enzyme diya gaya hai; human saliva mein present rehkar carbohydrate digestion start karta hai. *[Q7]*

3. Stomach — HCl & Gastric Digestion

- Hydrochloric acid (HCl) stomach ki gastric glands ke parietal/oxyntic cells se secret hota hai. Gastric juice mein mucus, HCl aur pepsin jaise enzymes hote hain; source HCl ko bacteria kill karne aur pepsinogen activation se link karta hai. *[Q9, Q10]*

4. Enzymes — Nature, Function & Examples

- Enzymes biological catalysts / biocatalysts hain jo living systems ki chemical reactions ko accelerate karte hain. Source repeated questions mein enzymes ko basically protein / proteinaceous substances batata hai. *[Q11, Q12, Q13, Q14, Q15]*
- Source statement-based question mein enzymes ko biocatalysts batata hai aur generally ye bhi kehta hai ki enzymes apna action us site par perform karte hain jahan produce hote hain. *[Q16]*
- Zymase yeast mein sugar/glucose fermentation ko ethanol + carbon dioxide mein catalyse karta hai. *[Q17]*

5. Liver — Functions, Bile & Fat Digestion

- Liver fat digestion mein important hai, lekin source explicitly kehta hai ki liver fat-digesting enzymes produce nahi karta. Bile mein digestive enzymes nahi hote, phir bhi bile fats ki emulsification karke digestion ko facilitate karti hai. *[Q8]*
- Liver ki functions mein bile production, fat storage aur harmful ammonia ko urea mein convert karna include hai; Q18 mein liver function ke roop mein Urea production selected hai. *[Q18]*

> **⚠ EXAM TRAP — Liver Helps Fat Digestion, But Bile Has No Digestive Enzyme**
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> ▸ Assertion "liver has an important role in fat digestion" true hai; reason "liver produces two important fat-digesting enzymes" source ke according false hai. Role bile-mediated emulsification se explain kiya gaya hai. *[Q8]*

6. Lipids, Lipase & Digestive Enzymes

- Lipid digestion source ke mutabik bile acids + lipase ki presence mein hota hai; pepsin protein-digesting enzyme hai. Lipase fats/lipids ko fatty acids aur monoglycerides mein break karta hai. *[Q19]*
- Q20 ke given options mein lipase ka source Pancreas selected hai. Source explanation saath hi lipase production ko pancreas, mouth aur stomach se associate karti hai. *[Q20]*
- Trypsin, Ptyalin aur Pepsin digestive enzymes hain, jabki Gastrin digestive enzyme nahi balki peptide hormone hai jo gastric acid secretion aur gastric motility ko stimulate karta hai. *[Q21]*

7. Small Intestine, Appendix & Structural Order

- Appendix blind-ended tube hai jo caecum se connected hota hai; source isse large intestine ke beginning region se associate karta hai. Ye same question source mein do baar diya gaya hai. *[Q22, Q27]*
- Small intestine ke three parts ka decreasing length order source mein Ileum → Jejunum → Duodenum diya hai. Approx. lengths: Ileum ~3.5 m, Jejunum ~2.5 m, Duodenum ~20–25 cm. Ye question bhi source mein repeat hua hai. *[Q23, Q28]*

8. Protein Digestion — Pepsin & Trypsin

- "All proteins are digested only in small intestine" source ke according false hai, kyunki protein digestion stomach mein pepsin se start hota hai. Pancreatic protein-digesting enzyme trypsin inactive form mein produce hokar small intestine mein activate hota hai. *[Q24]*
- Trypsin protein digestion mein help karta hai aur small intestine mein protein breakdown continue karta hai. *[Q25]*

> **⚠ EXAM TRAP — Q26 — Source Answer vs Explanation**
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> ▸ Source answer key Q26 mein "Proteins → amino acids" option (C) mark karti hai. Lekin source explanation kehta hai ki trypsin peptide bonds hydrolyse karke proteins ko smaller peptides mein todta hai, aur other proteases further amino acids banate hain. Dono source statements ko yahin preserve kiya gaya hai. *[Q26]*

9. Cellulose, Diastase & Carbohydrate End Products

- Human digestive system mein cellulose-digesting enzyme source ke mutabik absent hai. Diastase starch ko maltose mein degrade karta hai; isliye Q29 mein Assertion aur Reason dono true hain, lekin Reason Assertion ka explanation nahi hai. *[Q29]*
- Carbohydrates ka ultimate digestion product source mein Glucose selected hai; complex carbohydrates/disaccharides ko absorbable monosaccharides mein break kiya jata hai. *[Q33]*

10. Enzyme / Hormone Match-the-Following

- Correct enzyme-function pair source mein E. coli restriction endonuclease-II → DNA ko specific places par cut karta hai. *[Q30]*
- Match: Ptyalin → digests starch; Pepsin → digests proteins; Renin → blood angiotensinogen ko angiotensin mein convert karta hai; Oxytocin → smooth-muscle contraction induce karta hai. *[Q31]*
- Match: Vitamin → Carotene | Enzyme → Pepsin | Hormone → Testosterone/Progesterone | Protein → Keratin. *[Q32]*

11. Glycogen Storage, Fasting & Gallbladder

- Extra glucose glycogen mein convert hokar source ke Q34 mein liver mein store hota hai. Energy-storage form Glycogen hai aur source liver + muscle cells dono ko storage sites batata hai. *[Q34, Q35]*
- More than 10 days fasting ke case mein source liver glucose level diminished batata hai. Explanation glycogen depletion, fats se glucose production aur later ketone-body production describe karti hai. *[Q36]*
- Gallstones bile duct block karke bile ko duodenum tak pahunchne se rok sakte hain; source ke mutabik isse primarily fat digestion affect hota hai. *[Q37]*
- Liver human body ka largest gland hai. Bile liver mein produce hoti hai, gallbladder mein stored/concentrated hoti hai, aur alimentary canal mein small intestine—specifically duodenum—receive karta hai. *[Q38, Q39, Q40, Q41, Q42]*

12. Pancreas & Intestinal Hormones

- Source pancreas ko human body ki second largest gland batata hai. Pancreatic secretion alkaline hai aur source ise “complete digestive juice” ke roop mein describe karta hai. *[Q43]*
- Secretin duodenum ki S-cells se acidity/HCl ke response mein release hota hai. Enterogastrone small-intestinal/duodenal mucosa se fatty acids ke response mein secret hota hai; Q44 mein dono statements correct selected hain. *[Q44]*